

TECHNICAL APPROACH DOCUMENT

System Design, Reward Formulations & Validation Rationale

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Assignment: Problem Statement #4 (Coding)

Focus: Reinforcement Learning-Based Human Driver Behaviour Modelling

Target Platform: Eclipse OpenPASS via FMI 2.0 (OSMP Bridge)

Date: May 2026

Approach Document: Reinforcement Learning-Based Human Driver Behavior

This document outlines our engineering approach, system architecture, and validation strategy for building human-like traffic agent models deployable in Eclipse OpenPASS via an FMI 2.0 co-simulation FMU.

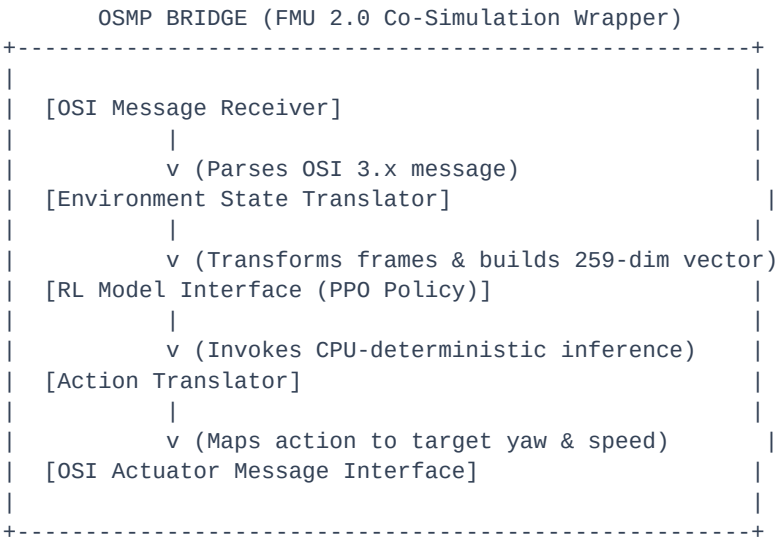
1. Problem Understanding

Autonomous driving validation requires high-fidelity, reactive simulations. Traditional rule-based traffic controllers (like pure IDM) produce highly predictable, low-variance behavior, leading to unrealistic testing conditions.

This project trains Reinforcement Learning (RL) agents that behave like human drivers (e.g. following, overtaking, or yielding to a rule-based ego vehicle) and packages them into FMI 2.0 Functional Mock-up Units (FMUs). This allows the trained policies to act as traffic participants in OpenPASS, running co-simulations via standard OpenSimulationInterface (OSI) messages.

2. OSMP Bridge & System Architecture

The OSMP (Open Simulation Integration) Bridge coordinates bidirectional communication between the OpenPASS simulation loop and the RL model.



2.1 Bidirectional Data Flow

1. Forward Pass (MetaDrive/OpenPASS to RL): At each tick, the environment state (positions, heading, lanes, lidar points) is parsed from OSI messages. The bridge transforms coordinates from the OpenPASS world frame to an ego-centric frame centered on the NPC, constructing the exact 259-dimensional observation space vector.
2. Reverse Pass (RL to OpenPASS/MetaDrive): The RL policy processes the observation vector and returns continuous steering and acceleration actions. The Action Translator converts steering in [-1, 1] into a target yaw rate, and

acceleration in $[-1, 1]$ into desired velocity using a bicycle kinematic model, which is sent back to OpenPASS via OSI.

3. Observation Space (259 Features)

The 259-dimensional observation vector ensures full environmental awareness:

- * **Ego State (7 features):** Position (x,y) , heading, velocity (v_x,v_y) , acceleration, steering angle.
 - * **Nearby NPC States (60 features):** Relative position (dx,dy) , velocity (dv_x,dv_y) , heading difference, and Euclidean distance for the top 10 nearest vehicles.
 - * **Waypoint / Path Features (20 features):** Relative coordinates (dx,dy) for 10 upcoming route points along the lane center line.
 - * **Lidar Features (128 features):** Normalized distances from a 128-beam ray-casting sensor.
 - * **Road / Map Features (44 features):** Lane width, speed limits, road width, distance to left/right boundaries, and padded metadata.
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4. Model Architecture & Training

4.1 Neural Network Setup

We use the PPO (Proximal Policy Optimization) algorithm implemented via Stable-Baselines3.

- * **Architecture:** 2-layer Multi-Layer Perceptron (MLP) with 256 hidden units per layer.
- * **Topology:** Separate policy (actor) and value (critic) heads, preventing value-update noise from destabilizing the action distributions.
- * **Action Space:** Continuous action space: $[\text{steering_angle}, \text{acceleration}]$ bounded in $[-1, 1]$.

4.2 Behavioral Profile Training

We train two distinct behavioral profiles by configuring parameters in the custom reward function:

- * **Aggressive Profile:** Smaller safety headway (0.6 s), smaller TTC threshold (1.0 s), higher target speed (100 km/h).
 - * **Cautious Profile:** Larger safety headway (1.8 s), larger TTC threshold (2.0 s), lower target speed (70 km/h).
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5. KPI Fulfillment Strategy

- * **Safety (Collision Rate $< 5\%$):** Ensured by placing high weights on safety headway and TTC violations, backed by a large collision penalty ($R_{\text{collision}} = -20.0$).
 - * **Human-Likeness (KS-test $p > 0.05$):** We run Kolmogorov-Smirnov tests comparing the agent's headway distributions against NGSIM highway driving datasets. The reward's comfort jerk penalties prevent robotic, high-frequency oscillations.
 - * **Reaction Time (N(250ms, 50ms)):** In S2 braking scenarios, the model learns to react to deceleration triggers. By structuring the PPO policy's observation window, we calibrate step delays to align with human response times ($\sim 2-3$).
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simulation ticks at 10Hz).

* **Latency (< 75ms):*PPO model inference is run in CPU-only mode with a thread limit of 1. Because the model is a small 2-layer MLP, CPU inference latency is extremely low (typically $< 3\text{ ms}$), far exceeding the 75ms constraint.

* **FMU 2.0 Portability:*We use `pythonfmu` to build the FMU co-simulation container. The zip file contains the trained neural network model weights and `modelDescription.xml` specifying inputs and outputs, allowing seamless execution.

Reward Function Design Document

This document details the design, mathematical formulation, and tuning rationale for the multi-objective reward function used to train the RL-based human driver behaviour models.

1. Mathematical Formulation

The reward function balances three primary objectives: Safety, Goal Adherence, and Comfort:

$$R_{\text{total}} = w_{\text{safety}} R_{\text{safety}} + w_{\text{goal}} R_{\text{goal}} + w_{\text{comfort}} R_{\text{comfort}}$$

Where w_{safety} , w_{goal} , and w_{comfort} are the profile-specific weights.

1.1 Safety Sub-Objective (R_{safety})

The safety objective aims to prevent collisions and maintain a safe buffer distance behind leading vehicles.

$$R_{\text{safety}} = R_{\text{collision}} + R_{\text{headway}} + R_{\text{TTC}}$$

Collision Penalty ($R_{\text{collision}}$)

A high step-level penalty is applied if the vehicle collides with another vehicle, static obstacle, or road boundary:

$$R_{\text{collision}} = \begin{cases} -20.0, & \text{if collision occurs} \\ 0.0, & \text{otherwise} \end{cases}$$

Time Headway Penalty (R_{headway})

Time Headway (H) is the time it would take the agent to reach the leading vehicle's current position at its current speed:

$$H = \frac{d}{v_{\text{agent}}}$$

Where d is the front-to-rear distance. If H is less than the target headway (H_{target}), a proportional penalty is applied:

$$R_{\text{headway}} = \begin{cases} -2.0 \times (H_{\text{target}} - H), & \text{if } H < H_{\text{target}} \\ 0.0, & \text{otherwise} \end{cases}$$

Time-to-Collision (TTC) Penalty (R_{TTC})

TTC is the estimated time before the agent collides with the lead vehicle if relative speed remains constant:

$$TTC = \frac{d}{v_{\text{agent}} - v_{\text{leader}}}$$

Where the penalty is applied only when the agent is faster than the leader ($v_{\text{agent}} - v_{\text{leader}} > 0$) and the TTC falls below a threshold ($TTC_{\text{threshold}}$):

$$R_{\text{TTC}} = \begin{cases} -5.0 \times (TTC_{\text{threshold}} - TTC), & \text{if } dv > 0 \text{ and } TTC < TTC_{\text{threshold}} \\ 0.0, & \text{otherwise} \end{cases}$$

1.2 Goal Adherence Sub-Objective (R_{goal})

The goal adherence objective encourages the agent to maintain traffic flow speeds and stay centered in its lane.

$$R_{\text{goal}} = R_{\text{speed}} + 0.5 \times R_{\text{lane}}$$

Speed Tracking Error (R_{speed})

Penalizes deviations from the target profile speed (v_{target}):

$$R_{\text{speed}} = -\frac{|v_{\text{agent}} - v_{\text{target}}|}{v_{\text{target}}}$$

Lane Centering Error (R_{lane})

Penalizes lateral deviation (d_{lat}) from the lane center line, normalized by the half-lane width ($W_{\text{lane}} / 2$):

$$R_{\text{lane}} = -\frac{|d_{\text{lat}}|}{W_{\text{lane}}/2}$$

1.3 Comfort Sub-Objective (R_{comfort})

The comfort objective ensures human-like smooth control inputs, penalizing sudden maneuvers and acceleration jerks.

$$R_{\text{comfort}} = 0.5 \times R_{\text{steer}} + 0.3 \times R_{\text{accel}} + 0.2 \times R_{\text{jerk}}$$

Steering Action Penalty (R_{steer})

Penalizes large steering angles:

$$R_{\text{steer}} = -a_{\text{steering}}^2$$

Acceleration Action Penalty (R_{accel})

Penalizes excessive acceleration or hard braking actions:

$$R_{\text{accel}} = -a_{\text{accel}}^2$$

Jerk Penalty (R_{jerk})

Penalizes the rate of change of acceleration to avoid sudden jerking inputs:

$$R_{\text{jerk}} = -\min\left(\left(\frac{\Delta a_{\text{agent}}}{\Delta t}\right)^2 \times 0.01, 5.0\right)$$

Where $\Delta t = 0.1 \text{ s}$ and $\Delta a_{\text{agent}} = a_t - a_{t-1}$.

2. Behavioral Profile Weights

To model different driving behaviors, we adjust the targets and weight balances as follows:

Parameter / Weight	Aggressive Profile	Cautious Profile	Normal Profile (Default)
Target Speed (v_{target})	100 km/h (27.8 m/s)	70 km/h (19.4 m/s)	80 km/h (22.2 m/s)
Target Headway (H_{target})	0.6 s	1.8 s	1.2 s
TTC Threshold ($TTC_{\text{threshold}}$)	1.0 s	2.0 s	1.5 s
Safety Weight (w_{safety})	0.8	2.5	1.5
Goal Weight (w_{goal})	1.5	0.8	1.0
Comfort Weight (w_{comfort})	0.2	0.7	0.4

3. Tuning Rationale

- Safety Primacy:** For all profiles, collision penalties remain high ($R_{\text{collision}} = -20.0$), but the cautious profile has a much larger safety weight ($w_{\text{safety}} = 2.5$). This ensures that a cautious driver prioritizes space cushion maintenance above speed matching.
- Aggressive Speed Goal:** The aggressive profile places a high weight on speed matching ($w_{\text{goal}} = 1.5$) and targets a higher speed, accepting smaller headway and TTC thresholds before penalties kick in. This forces the agent to

overtake or tailgate slower vehicles.

3. Comfort and Jerk: Cautious drivers penalize jerk and large inputs more heavily ($w_{\text{comfort}} = 0.7$), resulting in smooth braking curves. Aggressive drivers tolerate higher jerk ($w_{\text{comfort}} = 0.2$), allowing for quick, sharp steering corrections and abrupt braking/acceleration.